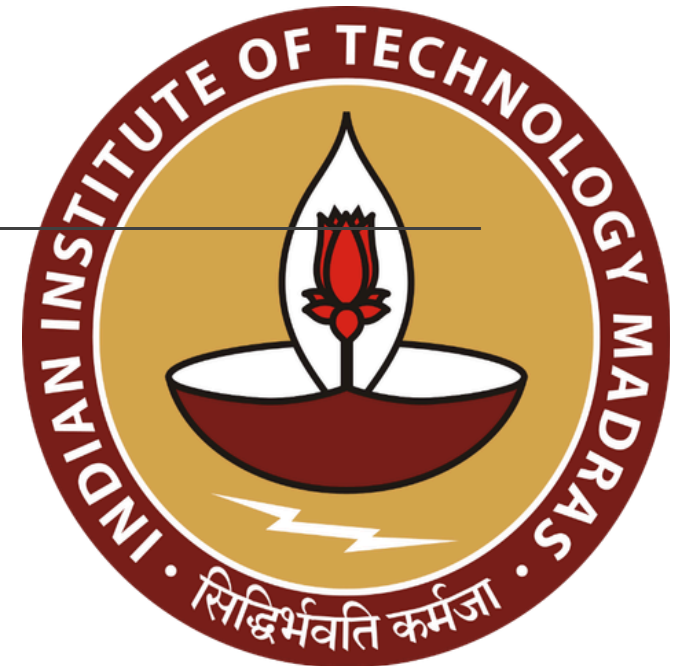




Strategic Growth and Revenue Stabilization: Enhancing Customer Retention and Inventory Management

NAME : SINAN AMEER S H
ROLL NO : 21F3000581
BUSINESS DATA
MANAGEMENT PROJECT



Organization Background

'S' and 'S' Brothers, a key wholesale dealer in jaggery, is located in Kozhikode Big Bazaar in Kerala. The business deals in B2B as well as B2C in the segment of 'Food and Grains' with an experience of more than 8 years.

The owner has over 25 years of experience in the jaggery trade. Before starting this venture, he was in a partnership business with his older brothers in the same industry.

'S' and 'S' Brothers offers high-quality, FSSAI-certified jaggery products, ensuring safety and compliance with food standards.

The venture has struggled with sharp revenue fluctuations due to inconsistent pricing and inefficient stock forecasting.



Problem 1 : Revenue Instability

- due to price fluctuations and increasing customer churn

Problem 3 : Weak Marketing

- Intense market competition forces frequent price adjustments, causing price-sensitive customers to defect to lower-cost competitors.

Problem 2 : Inefficient Inventory management

- Poor stock tracking causes frequent stockouts of high-demand items and overstocks of slow-moving goods, trapping cash flow and hurting profitability.

Problem 4 : Quality Issues

- Inconsistent jagged quality damages customer trust and reduces long-term retention.
- Use of artificial colors and excess sugar for visual appeal and shelf-life undermines product integrity.

Data Collection

Primary Data

- Stock Ledgers
- Sales Registers
- Bill book
- Day book

The raw data comprised transaction-level sales and inventory logs for a year(April 2023 – March 2024). Initial steps involved consolidating daily records into a master dataset. The data sources can be accessed

through the link mentioned below:

Stock Ledger

Competitor Pricing data was also collected through direct market intelligence.

Excel and Python (Google Colab) were used for data cleaning and initial analysis.

STOCK REGISTER											
Name of Article: B. H. 005 JH. 1. 1. 0000585											
DATE	PARTICULARS	V. No. of Bill No.	RECEIPT			ISSUED			BALANCE		RATE
			Quantity	Amount	Rs. p.	Quantity	Amount	Rs. p.	Quantity	Amount	
APRIL 01	OPSTOCK					74					
APRIL 02	BY SALES					73	21		73		
APRIL 03	OPSTOCK										
APRIL 04	BY SALES					52	21		52		
APRIL 05	OPSTOCK										
APRIL 06	BY SALES	273	48	56450		48					
APRIL 07	OPSTOCK					13			87		
APRIL 08	BY SALES					27	20		65		
APRIL 09	OPSTOCK					65					
APRIL 10	BY SALES					17			48		
APRIL 11	OPSTOCK										
APRIL 12	BY SALES	409	26	27240		48					
APRIL 13	OPSTOCK					74			73		
APRIL 14	BY SALES										
APRIL 15	OPSTOCK					53	20		53		
APRIL 16	BY SALES					17			34		
APRIL 17	OPSTOCK					34					
APRIL 18	BY SALES	415	98	128410		91			91		
APRIL 19	OPSTOCK										
APRIL 20	BY SALES					09	20		09		
APRIL 21	OPSTOCK					09			NIL		
APRIL 22	BY SALES	427	64	72100		64					
APRIL 23	OPSTOCK										
APRIL 24	BY SALES	434	60	77240		40			42		
APRIL 25	OPSTOCK					60					
APRIL 26	BY SALES					17			87		
APRIL 27	OPSTOCK					27	20		60		
APRIL 28	BY SALES					60					

BHARATHI TRADERS - Jaggery Nuts											
April		Opening Stock	By Sales	Incoming Stock	Total Stock	Revenue Generated		Unit Price	Total revenue		
1	2023	94	21	0	73	25515					
2	2023	73	21	0	52	25515					
3	2023	52	0	0	0	0					
4	2023	52	13	48	87	15795					
5	2023	87	22	0	65	26730					
6	2023	65	17	0	48	20655					
7	2023	48	1	26	73	1215					
8	2023	73	20	0	53	25800					
9	2023	53	19	0	34	24510					
10	2023	34	0	0	34	0					
11	2023	34	41	98	91	80490					
12	2023	91	82	0	9	125460					
13	2023	9	9	0	0	13770					
14	2023	0	0	0	0	0					
15	2023	0	0	0	0	0					
16	2023	0	22	64	42	29040					
17	2023	42	0	0	0	0					
18	2023	42	17	62	87	22440					
19	2023	87	25	0	62	33000					
20	2023	62	23	0	39	31395					
21	2023	39	24	0	15	32760					
22	2023	15	13	30	32	17745					
23	2023	32	14	0	18	18590					
24	2023	18	0	0	18	23760					
25	2023	18	18	0	0	23760					
26	2023	0	0	0	0	0					
27	2023	0	0	0	0	0					
28	2023	0	0	0	0	0					
29	2023	0	0	32	32	0					
30	2023	32	0	0	32	0					
		TOTAL SALES		TOTAL INCOMING		OPENING STOCK					
		422		360		94					
IN KILOGRAMS(30KG)		12660		10800		2820		1464.32		61794	

Methodology Used:

01

Descriptive Statistics

- Calculated mean, median, SD, minimum and maximum for sales, revenue and price.
- Helps identify top products, revenue patterns and sales variability.
- eg :Bharati Traders(Jaggery Nuts): has a high mean but also high std and months with zero sales, indicating risk and volatility

03

ABC Categorization

- Classified SKUs based on their sales/revenue contribution.
- A: Top 70–80% | B: Next 15–20% | C: Bottom 5–10%.
- Helps prioritize high-value products.

02

Time Series Analysis

- Analyzed sales, revenue and prices over time.
- Identified peak and low-demand periods to support better inventory and promotional decisions.

04

Competitor Pricing Analysis

- Compared competitor prices for key products with our average selling prices.
- Helps assess pricing competitiveness.

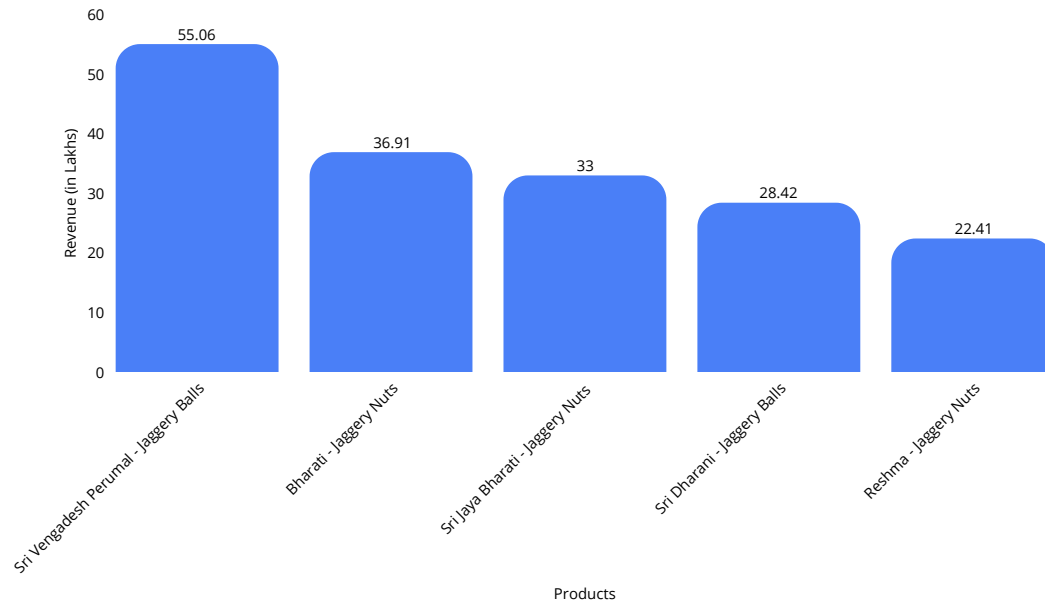
05

Stock Return Analysis

- Analyzed the quantity of products returned by customers.
- Helps identify quality issues and reduce product returns.

Findings – ABC Categorization

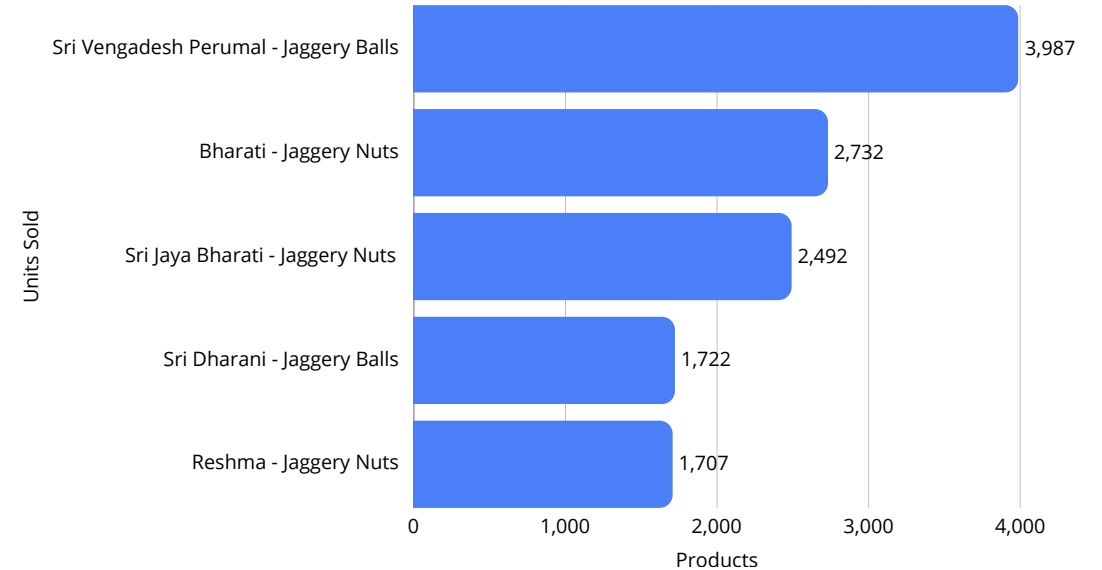
ABC Categorization on Total Revenue



“A” category SKUs (Sri Vengadesh – Jaggery balls, Bharati – Jaggery nuts) contribute over 50% of total revenue and sales.

“B” category SKUs (SJB – jaggery nuts, Sri Dharani – Jaggery balls) add significant but moderate value. These require the most stringent inventory control, price monitoring, and promotion focus.

ABC Categorization based on Sales

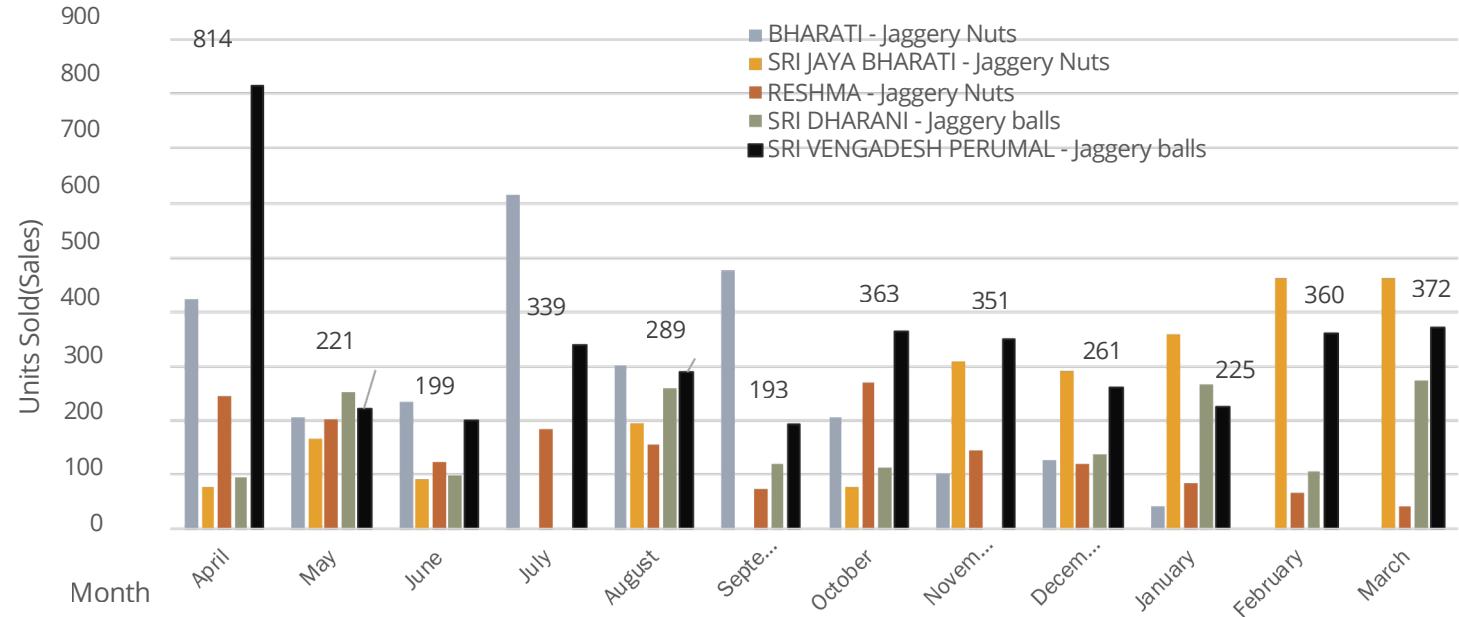


Sri Vengadesh Perumal – Jaggery balls contributes highest in terms of revenue as well as sales, contributing ₹55,06,300 to the revenue and 3987 in sales volume.

“C” category SKU (Reshma – Jaggery nuts) has minimal impact, represents slow movers. Reshma traders – Jaggery nuts has minimal impact in revenue and sales, potential candidate for inventory reduction or clearance.

Findings – Sales & Price Trends

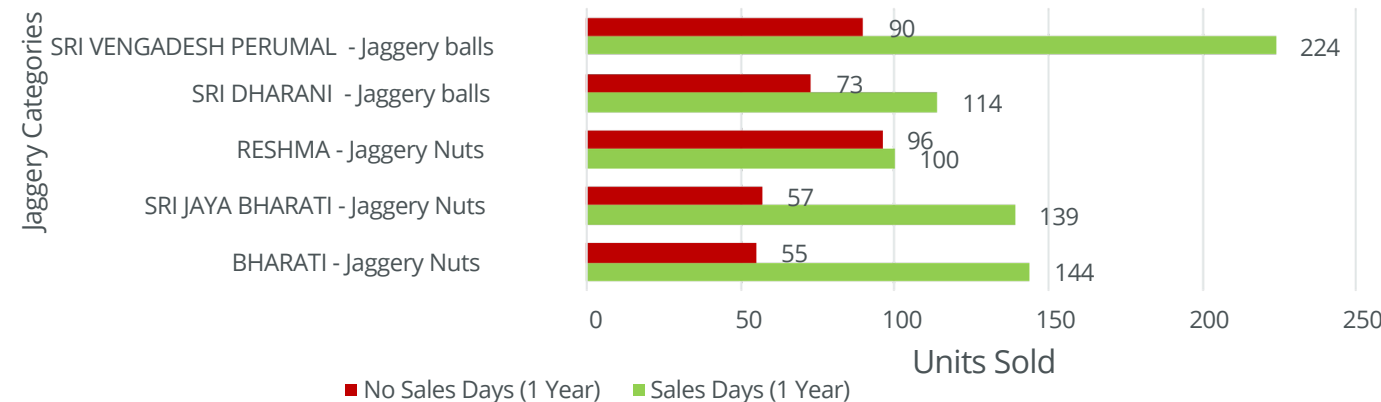
Sales Trend For Each Items



Sales volumes for the major SKUs show clear seasonality and volatility:
Example: Bharati Traders – Jaggery Nuts experienced a dramatic sales spike in July but then sales declined sharply, reaching zero in February and March, even when there was available stock.

Sri Jaya Bharati Traders and others also show irregular sales, with months of high peaks and others with little or no activity.

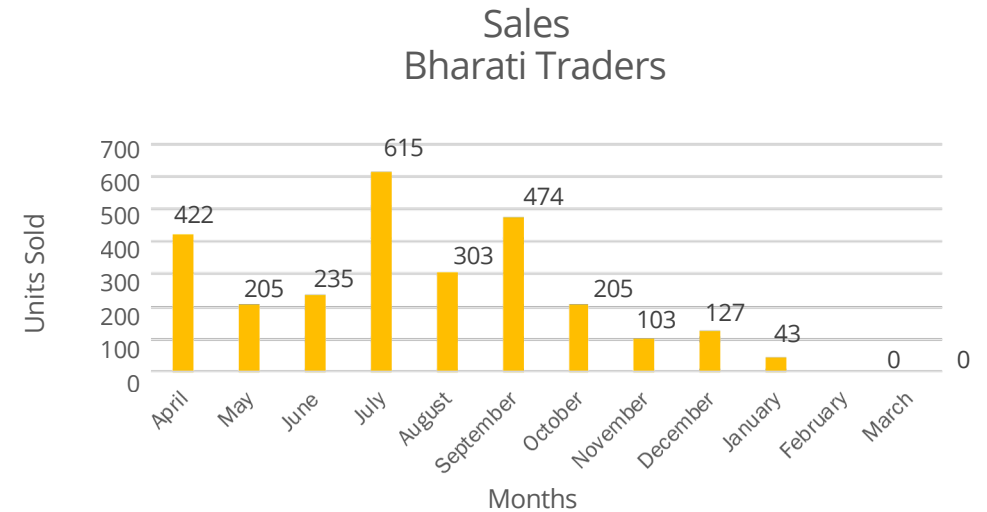
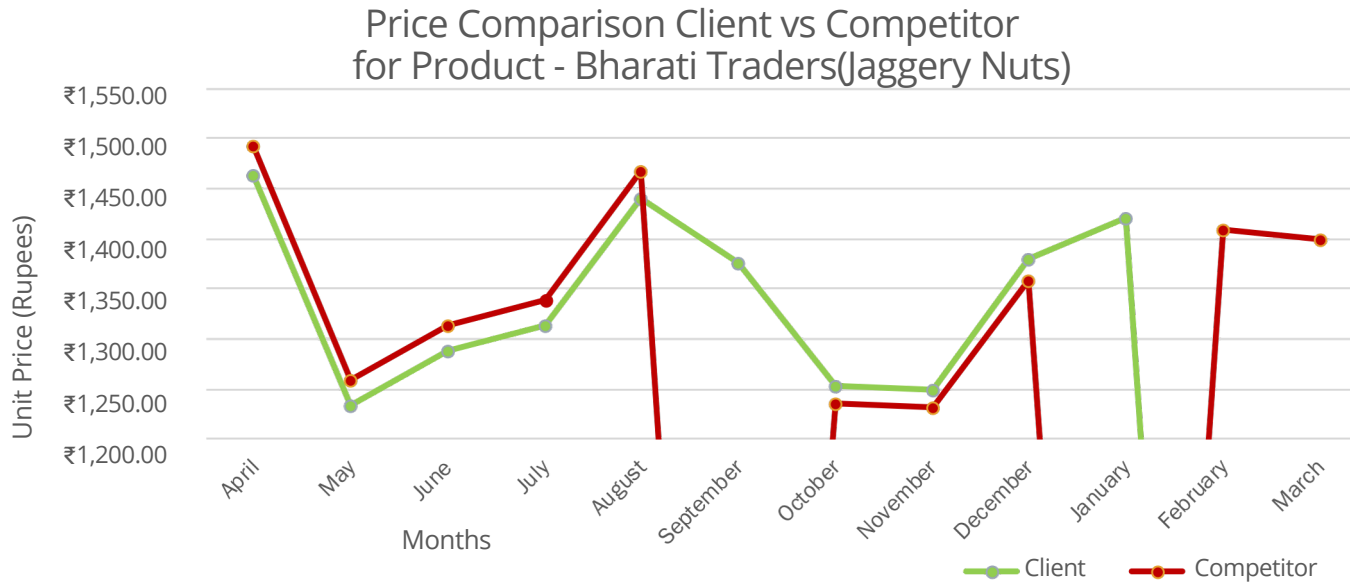
Sales To No Sales Day for Each Product



The number of “no sales days” is a critical operational issue for all products, particularly some SKUs that remain unsold for large parts of the year. Focusing on understanding the seasonality patterns and minimizing no-sales days is essential for stabilizing revenue and improving stock turnover.

Findings – Competitive Pricing

Based on CompetitorPricing: Example BHARATI–Jaggerynuts



These show “Price Comparison” between client vs competitor for Bharati – jaggery nuts. Similarly, the comparison is also available for SJB – Jaggery nuts. When the client prices align with or undercuts the competitor, sales volumes are higher. These figures best illustrate the direct relationship between pricing position and sales volume, making the price sensitivity clear. Even small price differences (2-3%) cause sharp sales drop, confirming high price sensitivity in the market. Tight alignment with competitor pricing is critical for defending and growing sales, hence customer retention.

Interpretations

1. Revenue & Sales Instability

- Key SKUs show high sales fluctuations and seasonality.
- Sales often peak during festivals and fall in other months.
- Eg: Bharathi and SJB - Jaggery Nuts

2. Inventory Challenges

- Frequent no-sales periods indicate overstocking.
- Better inventory planning is needed to improve cash flow.
- Eg: Sri Dharani - Jaggery balls and Reshma - Jaggery nuts

3. High Price Sensitivity

- Sales are highly affected by small price changes.
- Competitive pricing is important to maintain sales.

4. Product Quality Concerns

- High product returns indicate possible quality issues.
- This can affect customer trust and revenue.
- Eg: Reshma - Jaggery Nuts

Recommendations

Implement Dynamic Pricing

Adjust prices based on competitor pricing to maintain sales.

Focus on High-Impact SKUs (ABC Analysis) :

Prioritize A & B products for marketing, inventory and pricing. Reduce excess stock of C products.

Adopt Data-Driven Demand Forecasting:

- Use past sales and seasonality to plan stock and avoid stock-outs.
- Automate replenishment reminders and set minimum stock thresholds for "A" SKUs

Strengthen Supplier Quality:

- Conduct batch-wise quality checks to reduce product returns.